MASTER OF ARCHITECTURE THESIS · SAVANNAH COLLEGE OF ART AND DESIGN

The Powerhouse

A sustainable charging anchor

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Savannah, Georgia · 2027

The Powerhouse

A Sustainable Charging Anchor

A Thesis Submitted to the Faculty of the Architecture Department in Partial Fulfillment of the Requirements for the Degree of Master of Architecture at Savannah College of Art and Design

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Dedications & Acknowledgments

[To be written.]

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Thesis Framework

HYPOTHESIS

An Electric Vehicle Powerhouse complex will turn charging activities into an opportunity for enhanced civic / community engagements.

CHAPTER 1 · INTRODUCTION

Background

Electrification and speed · the architecture of the stop · charge time · public space as community · the anchor · theoretical framework

SUSTAINABILITY

CHAPTER 2

The Stop

Parking: the paved ground
The charging network
Stations and service areas
Vehicle and battery size
Driver rest rules
Charger standards and zoning
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Right to the city (Lefebvre)
Who the space is for
Rent, revenue, running costs
Lessons from malls and hubs
Programme by length of stay
Surveillance and staying (Foucault)
Open to all

MEASURE: PROGRAMME ON SITE

PRECEDENT STUDIES AND TEST

Compare existing sites

Gas stations · service areas · truck stops · charging hubs, in corridor, city and town settings

FINDINGS

Site-specific design criteria

MODULE · ONE IS DESIGNED

City, highway or town

City: gym, workspace · Highway: rooms, showers · Town: clinic, market

SITE

Site and community analysis

GRADUATE THESIS 01–02

Site-specific design

OUTCOME

The Powerhouse

FINAL CHAPTER

Conclusion

back to the hypothesis: does it hold?

leads to

loops back

core idea

Abstract

Road transport is going electric, from e-bikes to heavy trucks, and every electric vehicle has to stop to charge. That stop can last from a quarter of an hour to a whole night. Yet most charging sites are still car parks with a row of plugs, used by drivers and by no one else.

This thesis proposes the Powerhouse, a new type of stop built on ground that parking already covers. Like the railway station and the airport, it treats a break in the journey as a reason to build a place: chargers alongside a programme chosen for the neighbourhood, and public space open to everyone. Chapter 1 sets out the argument and Chapter 2 studies the stop. Later chapters turn to the public space, the programme, a test across existing sites, and the design.

Hypothesis

An Electric Vehicle Powerhouse complex will turn charging activities into an opportunity for enhanced civic / community engagements.

Every electric vehicle, from e-bikes to trucks, has to stop to charge. The Powerhouse makes that stop a community one. The charge brings people to the site, a programme chosen for the neighbourhood gives them a reason to stay, and around it sits public space open to everyone, whether they charge or not.

The community stop. How long people stay to charge or rest, set by the vehicle, the charger and the law.

The programme. What the neighbourhood lacks, built around the charge: rooms for drivers on a freight corridor, a gym in a city, a clinic in a town.

The public space. The share of the site anyone can use without buying anything.

The Powerhouse

A sustainable charging anchor

Architecture has always been shaped around people, and especially around the way they move. The coaching inn, the railway station, the bus depot, the airport and the highway service area were each built around a pause in a journey, and each was sized to the time that pause allowed. That pause is now changing. Fuel is getting more expensive, vehicles are learning to drive themselves, and road transport is turning to electricity. A car that once stopped for five minutes to fill its tank now stops for close to an hour to charge. Yet the places built for charging have hardly changed from the car parks and forecourts they replaced, and many drivers avoid them altogether by charging at home.

This thesis, the Powerhouse, proposes a sustainable charging cluster: an Electric Vehicle Powerhouse complex that will turn charging activities into an opportunity for enhanced civic / community engagements. It is not one building repeated along the highway. Each Powerhouse takes its programme from what the community around it lacks, and that programme is what brings people in; the charge is what lets them stay.

Nor is it only for electric vehicles. The Powerhouse is open to everyone, including people who arrive on foot, by bus or by bicycle.

Thousands of new charging sites will be built over the coming decades, and most are being planned as infrastructure and nothing more. This thesis asks how they could instead be woven into the everyday needs of the communities they sit in, and what it would take for those places to sustain themselves. It starts from a simple belief: that the energy transition is a chance to leave behind a better and greener landscape than the one we have now.

1.1 Electrification and Speed

Transport uses a great deal of energy and wastes most of it. Of every hundred tonnes of greenhouse gas the United States emits, twenty-eight come from moving people and goods, more than from any other sector.[1] Across the world, road vehicles added just over six billion tonnes of carbon dioxide to the air in 2024.[2] Much of the problem is the engine. Fill a petrol car and roughly seven-tenths of that energy leaves through the exhaust and the radiator as heat; only about three-tenths actually moves the car. An electric car turns the ratio around, putting close to nine-tenths of what is in its battery to work.[3] At the wheels, it makes the same journey on roughly a third of the energy. Even so, an electric car is only as clean as the grid behind it. Economists who priced the damage done by driving electric across the United States found a clear benefit in California and a net harm in coal-heavy North Dakota.[4] This is one reason the Powerhouse generates its own energy.

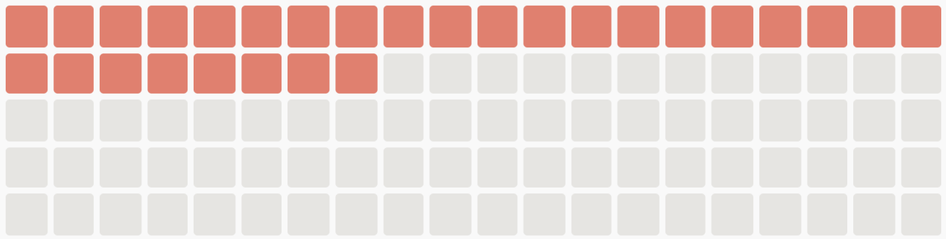
The shift is happening fast. Five years ago electric cars were a niche of a few million sales a year. In 2025 buyers took home more than twenty million.[5] Automation is arriving alongside it. In ten American cities, Waymo now sells around half a million rides a week in cars with nobody at the wheel,[6] and since 2025 Aurora's trucks have hauled paying freight between Dallas and Houston with the cab empty.[7] Cars have gone first. Freight and long-distance road transport are next. Speed is not the same as certainty, though. A review of ninety-four studies on electric-car adoption found that what actually persuades people to switch, public charging included, is far less settled than is usually assumed.[8]

For now, most charging happens at home. By the Department of Energy's estimate, four of every five charges happen in a driveway or a garage.[9] That convenience has a cost. An electric car driven a typical American distance uses roughly three to four thousand kilowatt-hours of electricity a year, about a third of what an average home already uses.[10] Home charging moves the cost of the transition onto household bills, and is not an option for people who live in apartments or park on the street. Public charging will have to carry more of the load, and the places built for it will matter.

This section moves from the US network to California, New York and Florida. The question is simple: **where could charging also support everyday community life?**

28%

of US greenhouse gas comes from transport

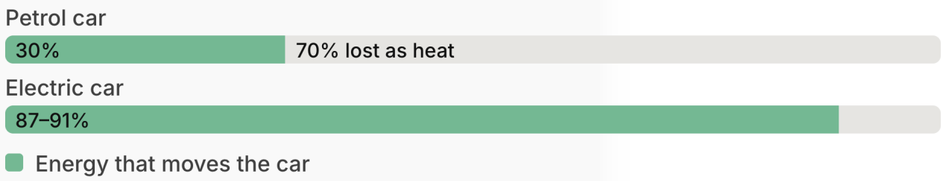


Transport

Source: EPA, Inventory of U.S. Greenhouse Gas Emissions and Sinks 1990–2022 (2024) · Note 1

3 in 10

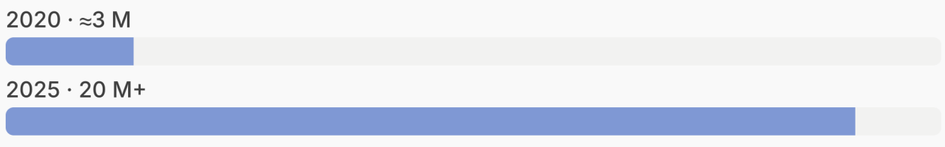
units of petrol move the car; an EV uses 9 in 10



Source: US DOE Vehicle Technologies Office, Fact of the Week #1360 (2024) · Note 3

20M+

electric cars sold worldwide in 2025



Source: IEA, Global EV Outlook 2025 and 2026 · Note 5

4 in 5

charges happen at home



Source: US DOE, National EV Charging Network · Note 9

33–38%

more electricity a year for a home with one EV



Source: EIA, household electricity use 2022; DOE/EPA fueleconomy.gov; author's estimate · Note 10

6 billion

tonnes of CO_2 from road vehicles worldwide, 2024



1 block = 1 billion tonnes

500k

driverless Waymo rides a week

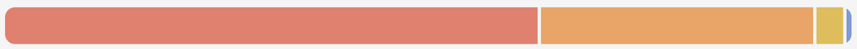


1 block = 100,000 rides

Sources: IEA, Breakthrough Agenda Report 2025 · Note 2; TechCrunch, "Waymo's Skyrocketing Ridership in One Chart" (March 2026) · Note 6

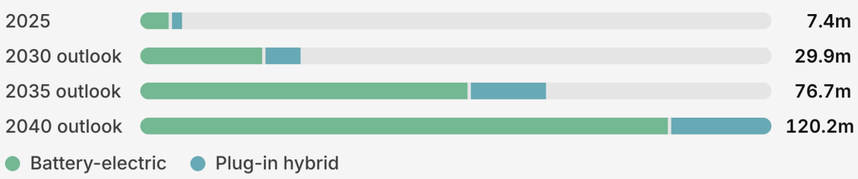
Where public charging is today

All vehicles · US · 2024 **297.5m**



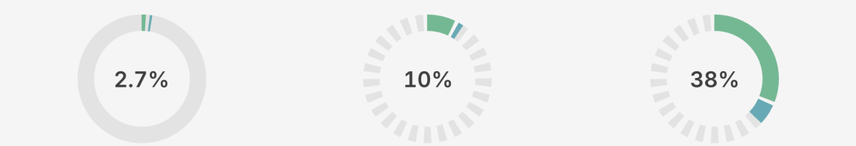
Trucks (incl. SUVs, pickups) 64% · Cars 33% · Motorcycles 3.1% · Buses 0.4%

Electric vehicles · US · 2025–2040 **7.4m → 120.2m**



Battery-electric · Plug-in hybrid

Share of US vehicles that are electric

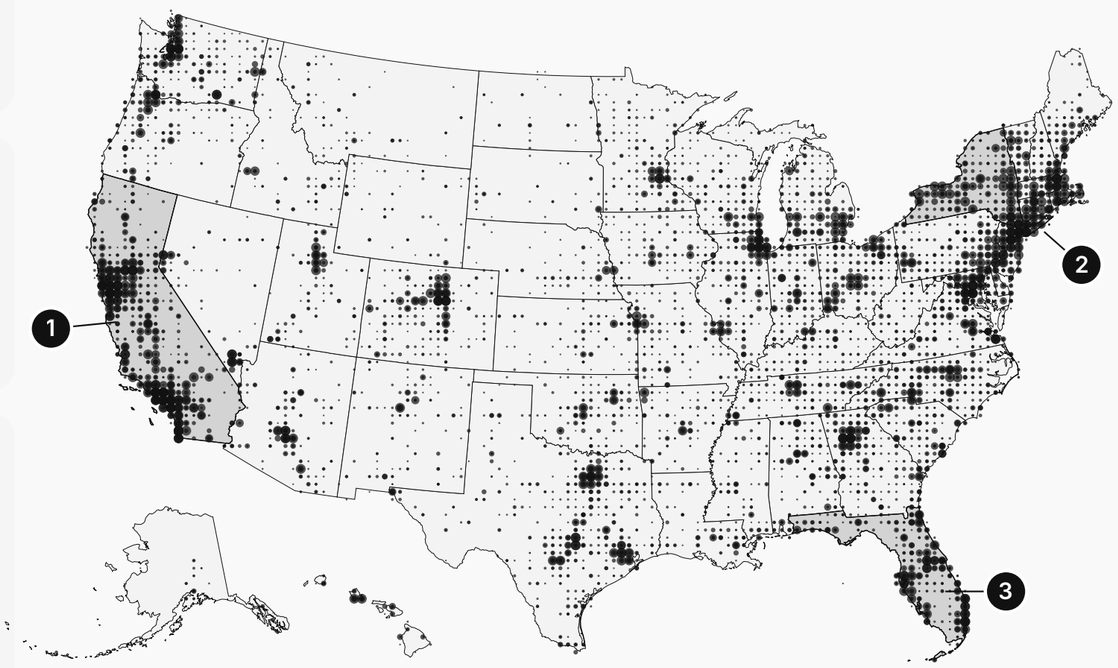


Battery-electric · Plug-in hybrid

All vehicles: FHWA Highway Statistics 2024, Table MV-1. Electric vehicles and share: US cars and light trucks, EIA Annual Energy Outlook 2026; 2025 is EIA's base year, later years are its baseline outlook, not a forecast.

Where the US's 290,019 charging points are · at 88,068 stations · public and private

Each dot on the map groups nearby charging locations, and its size grows with the number of charging points there. Public means open to any driver, even when the equipment is privately owned; private means workplaces, fleets and apartment blocks that report their chargers. Home chargers are not counted anywhere.



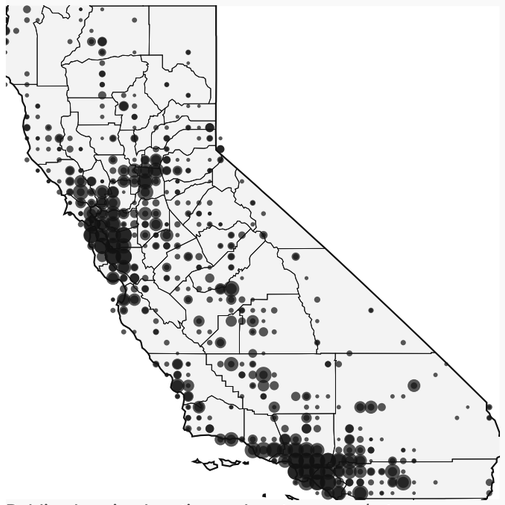
All 50 states and DC — **260,991 public charging points**

1	**California** 26.5%		69,255
2	**New York** 7.7%		20,184
3	**Florida** 5.9%		15,379
4	**Texas** 5.0%		13,004
5	**Massachusetts** 4.6%		11,891
6	**Washington** 3.5%		9,096
7	**Colorado** 3.0%		7,940
8	**Georgia** 2.9%		7,640

Three states, three different conditions

California, New York and Florida hold 40.2% of US public charging ports. Comparing these leading states shows three different conditions; it does not identify the country's greatest unmet need.

California

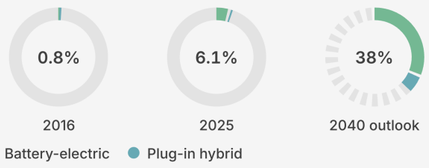


Public charging locations · drag to pan · +/− to zoom

All vehicles · 2024 **30.9m**

Trucks (incl. SUVs, pickups) 55% · Cars 42% · Motorcycles 2.4% · Buses 0.3%

Share of vehicles that are electric



Battery-electric · Plug-in hybrid

Charging infrastructure · Oct 2026 **79,787**

Public fast 25% · Public slow 62% · Private 13%

Electric vehicles · 2025 **2,334,000**

Public charging points · Oct 2026 **69,255**

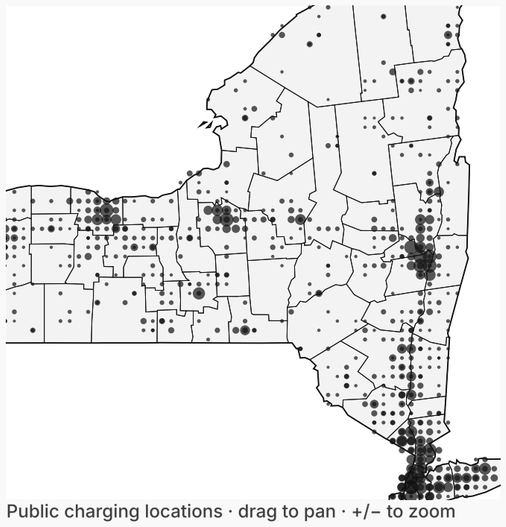
Electric vehicles per public charging point **34**

US average 28 (marker)

Battery-electric growth · 2024–25 **+20.2%**

The largest existing market of the three. Investigate gaps within a substantial charging network.

New York

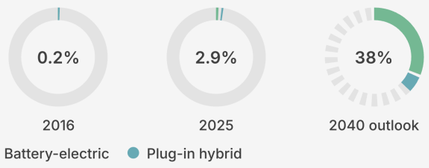


Public charging locations · drag to pan · +/− to zoom

All vehicles · 2015 **10.6m**

Trucks (incl. SUVs, pickups) 51% · Cars 46% · Motorcycles 3.3% · Buses 0.3%

Share of vehicles that are electric



Battery-electric · Plug-in hybrid

Charging infrastructure · Oct 2026 **21,531**

Public fast 15% · Public slow 78% · Private 6.3%

Electric vehicles · 2025 **334,400**

Public charging points · Oct 2026 **20,184**

Electric vehicles per public charging point **17**

US average 28 (marker)

Battery-electric growth · 2024–25 **+24.5%**

A smaller EV market, with a city where access to home charging is a particular challenge.

Florida

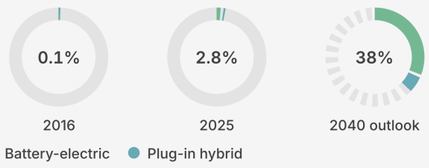


Public charging locations · drag to pan · +/− to zoom

All vehicles · 2024 **19.7m**

Trucks (incl. SUVs, pickups) 60% · Cars 37% · Motorcycles 3.2% · Buses 0.3%

Share of vehicles that are electric



Battery-electric · Plug-in hybrid

Charging infrastructure · Oct 2026 **16,840**

Public fast 32% · Public slow 59% · Private 8.7%

Electric vehicles · 2025 **538,800**

Public charging points · Oct 2026 **15,379**

Electric vehicles per public charging point **35**

US average 28 (marker)

Battery-electric growth · 2024–25 **+34.5%**

The fastest recent percentage growth of the three. Test where new demand requires shared charging.

Electric vehicles (EVs) here means battery-electric cars plus plug-in hybrids; ordinary hybrids that never plug in are counted as fuel vehicles. Growth measures battery EV registrations, not charger use. Vehicle and charger counts refer to different years.

Sources: AFDC registrations, 2016–25 · AFDC charging inventory, October 2026 · EIA AEO2026 national outlook, Table 39.

California, New York and Florida compared

Measure	California	New York	Florida
All vehicles FHWA; New York 2015, the others 2024	**30.9m**	10.6m	19.7m
Electric vehicles, 2025 battery-electric + plug-in hybrid	**2,334,000**	334,400	538,800
Electric share of cars and light trucks, 2025	**6.1%**	2.9%	2.8%
Battery-electric growth, 2016–2025	13×	34×	**39×**
Public charging points, Oct 2026 share of all US public charging points	**69,255** 26.5% of US	20,184 7.7% of US	15,379 5.9% of US
Fast chargers among public charging points a charge in 20–60 minutes	28%	16%	**35%**
Slow chargers among public charging points a charge of several hours: the long stop	72%	**84%**	65%
EVs per public charging point US average 28; higher means more drivers share each one	34	17 * Main reason · below the US average of 28	**35**
What it shows	Scale, and a cleaner grid	Where home charging is not an option	The fastest shift

Highest in each row in bold. Vehicle and charger counts come from different years; EVs per charging point is a ratio across them, not a measure of how busy the chargers are.
Sources: FHWA Highway Statistics, Table MV-1; AFDC vehicle registrations 2016–2025; AFDC station inventory, 18 Oct 2026.

Why New York

New York is where the Powerhouse's argument is sharpest. California has the scale and Florida the fastest growth, but the Powerhouse is not about the biggest market. It is about people who have to stop to charge and could stay, and New York's conditions make that stop most likely to happen and to last.

1. **Charging away from home is not a choice.** Half of New York City's car-owning households park on the street, with no driveway or garage to charge in. For them the one charge in five that most drivers make away from home is every charge.
2. **Its stops are long.** 84% of New York's public charging points are slow chargers, the highest share of the three (California 72%, Florida 65%). A charge takes hours, not minutes: time the Powerhouse turns into a place.
3. **The shift is early, and fast.** Battery-electric cars grew 34× since 2016, yet only 2.9% of vehicles are electric. Most of the charging the city will need is still to be built, and the city is already planning curbside chargers, so what gets built next can be designed as a place, not only as parking.

New York City is the test site for the Powerhouse. Its deep dive follows: where street-parked drivers live, where chargers are planned, and what each neighbourhood lacks.

Street parking: NYC DOT, curbside charging expansion, Aug 2026. Chargers: AFDC station inventory, 18 Oct 2026. Vehicles: AFDC registrations 2016–2025.

1.2 The Architecture of the Stop

Every kind of travel has produced its own architecture of waiting. On the Silk Road the caravanserai gave shelter, trade and exchange at the end of a day's march; along coaching roads the inn offered rest, food and news for travellers and the town alike. The great railway stations of the late nineteenth and early twentieth centuries were among the grandest civic buildings in their cities. Passengers arrived early and waited in vast halls, concourses and plazas, surrounded by restaurants, shops and newsstands. In New York, McKim, Mead and White borrowed the vaults of an ancient Roman bath for the waiting room of Pennsylvania Station (1910), and when Penn Station was demolished in 1963, New York lost its great civic room for the train and the city's preservation movement began.[11] Airports went further, turning the terminal into a city of waiting.

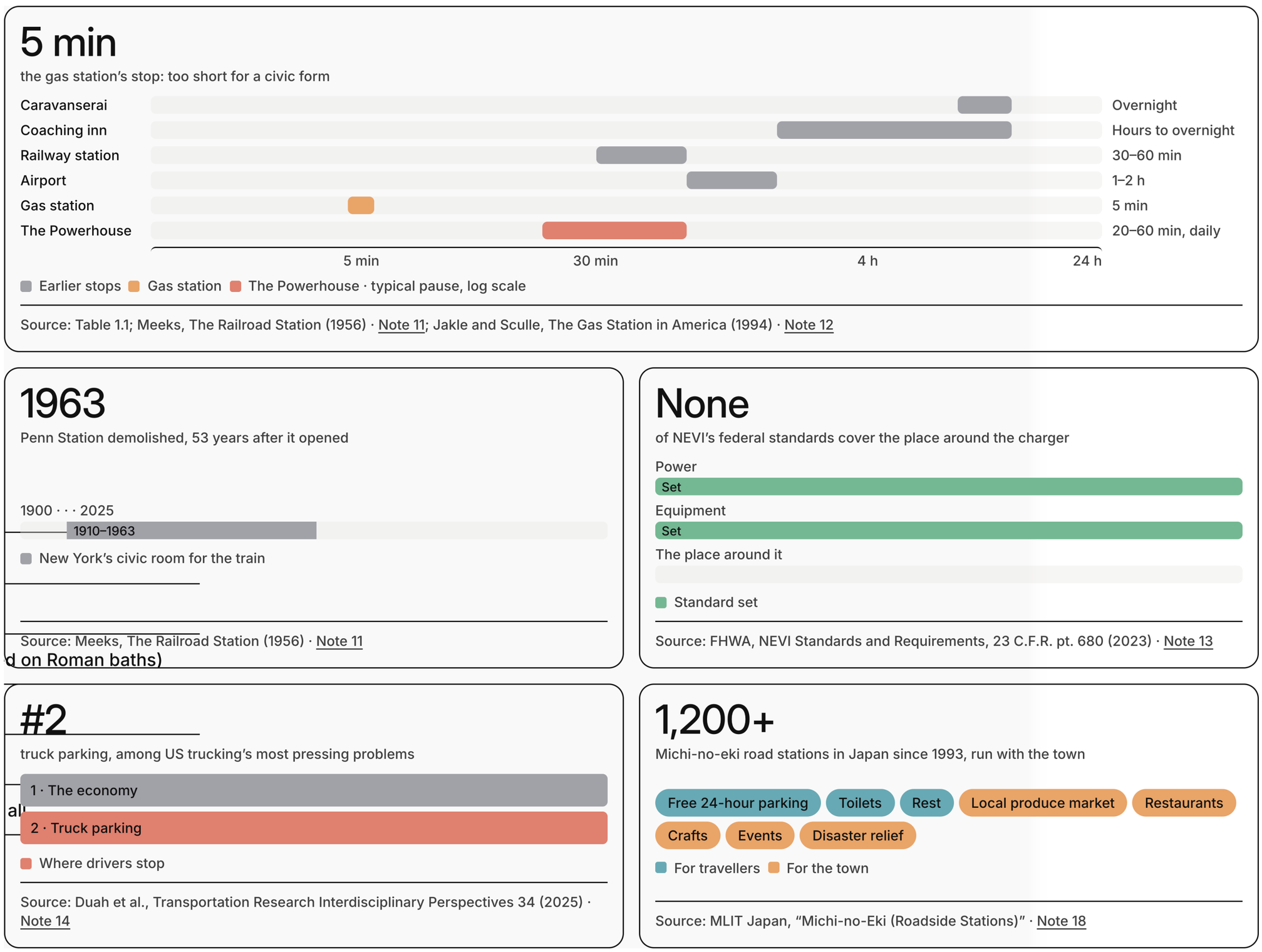
The roadside stop went the other way. For most of the twentieth century it meant the gas station, which settled early into a formula that hardly changed afterwards: a roof over the pumps, islands to pull up beside, and a booth where you paid on the way out.[12] It was built for grab-and-go, and it did that well. Five minutes is long enough to pay and too short for a civic form.

Table 1.1. Every age built a civic building around its stop: what is the charging stop's civic form?

Era & stop	Typical pause	Civic form & public life
Caravanserai	Overnight	Courtyard for shelter, trade and exchange on the Silk Road
Coaching inn	Hours to overnight	Rest, food and news for travellers and the town
Railway station	30–60 minutes	Civic hall and concourse (Penn Station, 1910: a waiting room modelled on Roman baths)
Airport	1–2 hours	The terminal as a self-contained city of waiting
Gas station	5 minutes	Forecourt canopy and payment booth; too short for a civic form
The Powerhouse	20 minutes to overnight, every day	Charging cluster, local energy and neighbourhood civic room open to all

The charging stop has inherited this form without its reason. The usual response is to add a few chargers to an existing car park. Public programmes such as the US National Electric Vehicle Infrastructure programme set standards for power and equipment, but say nothing about the place around the charger.[13] Drivers end up sitting in their cars beside a row of posts. Freight is worse off. American truckers already compete for somewhere to stop. Truck parking now ranks as the industry's second most pressing problem after the economy, and drivers near the end of their legal hours often keep driving in search of a space.[14] Very few charging sites are built for heavy vehicles. A recent review of the American trucking market names charging infrastructure, alongside range and upfront cost, as one of the main obstacles to electric freight.[15] As trucks electrify, their stops will get longer, with nowhere designed to hold them. Marc Augé's word for such spaces, non-places, fits them well: they are used all the time and belong to no one.[16]

A few operators have tried to do more. In England, Gridserve's forecourt at Braintree set a café, a shop and a lounge beside its chargers in 2020,[17] a glimpse of what a longer stop can support, though it is a commercial venture first. Japan has already proved the public version at national scale: since 1993 its Ministry of Land, Infrastructure, Transport and Tourism has designated more than 1,200 *Michi-no-eki* ("road stations"), each pairing free 24-hour parking, toilets and rest for travellers with a market for local produce, restaurants, crafts, events and disaster-relief facilities run with the municipality and local residents.[18]

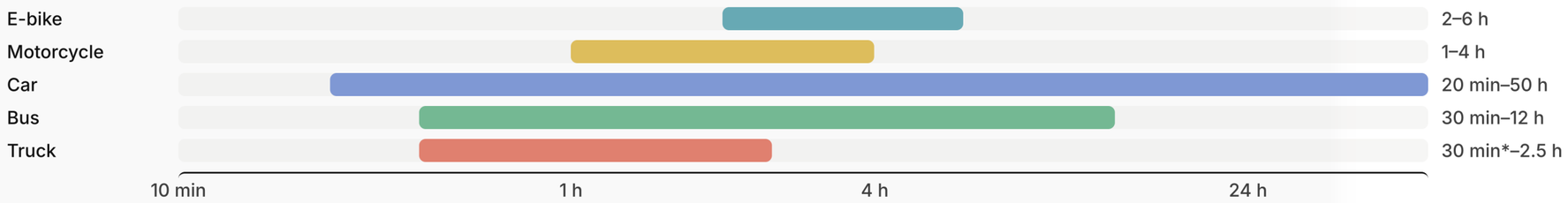


1.3 Charge Time

How long a vehicle stops depends on what it is and how it charges. A fast charge takes twenty minutes to an hour; a slower charger at home or at a bus depot takes several hours, usually overnight. For freight, the stop is set by law as well as by technology: American rules require half an hour off after eight hours at the wheel, and ten hours off duty at the end of a working day.[19] An electric truck can charge during a break the driver already has to take. The thesis looks at all road transport, not only cars. Trucks, buses, motorcycles, bicycles and e-bikes all need somewhere to stop and charge, and each brings different people to the site. The Powerhouse plans for all of these stays, and fits its programme to them.

20 min–50 h

to charge, from the fastest charger to the slowest, by vehicle size



Left end: fast charge, to 70–80% · right end: home or depot, to full · log scale · * Tesla Semi claim, unverified

Sources: Upway, “How Long Does It Take to Charge an Electric Bike?”; Zero Motorcycles, “Charging and Range”; US DOT, “Charger Types and Speeds”; Monta, “Electric Buses”; Tesla, Semi (2025); Freightliner, eCascadia spec sheet (2025); Volvo Trucks, VNR Electric

4–10 h

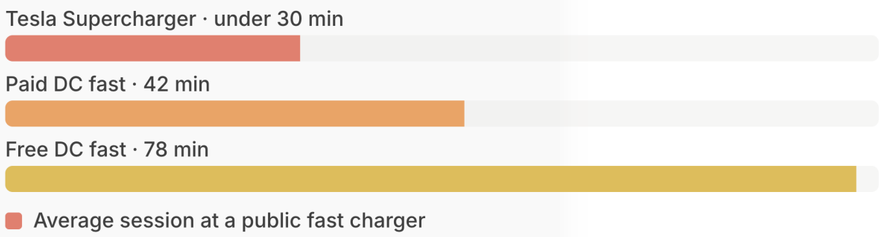
to fill a car at home on a Level 2 charger, overnight



Source: US DOT, “Charger Types and Speeds”

27 min

an average Tesla Supercharger stop, the kind sited by malls and restaurants



Sources: Tesla Charging, average Supercharging time 2017–2025 (Feb 2025); US DOE, Fact of the Week #1319 (Dec 2023), 2.4 million sessions

30 min

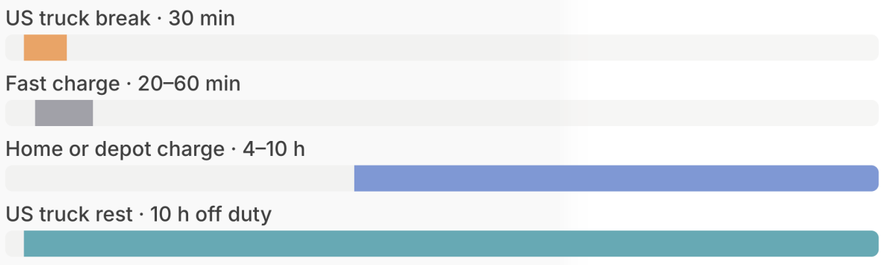
break a US truck driver must take after 8 hours at the wheel



Source: 49 C.F.R. § 395.3 · Note 19

20 min–10 h

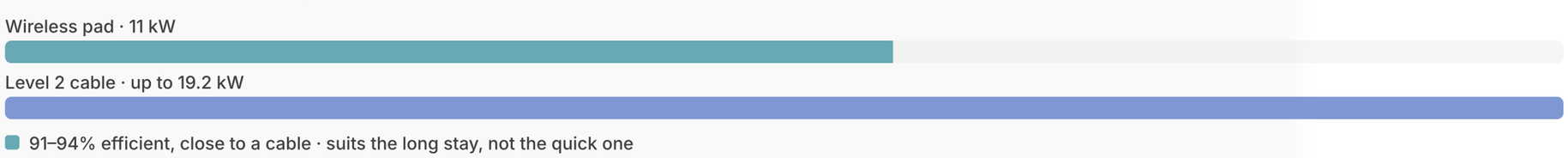
the stays the Powerhouse plans for, from a quick charge to an overnight rest



Sources: US DOT, “Charger Types and Speeds”; 49 C.F.R. § 395.3

11 kW

wireless: park over a pad and charge with no plug, at about Level 2 speed



Sources: Electreon, Lite DOT (2026); WiTricity, 11 kW system (2016); WSU Energy Program, charging speeds

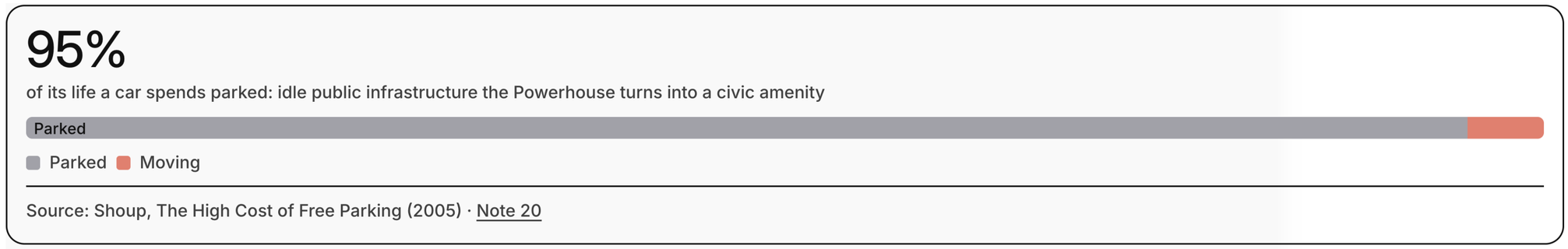
1.4 Public Space as Community

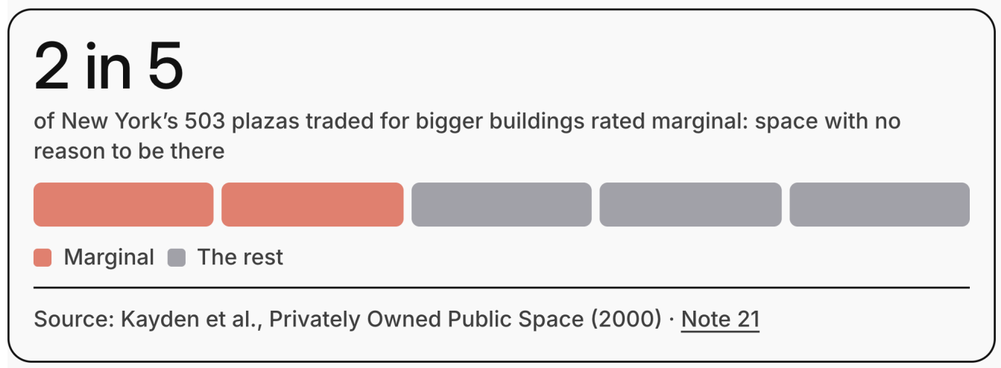
A long stay gives a site time, but not a reason to be there. Few kinds of land in America are as plentiful, or as idle, as the parking lot. The typical car stands still about 95 per cent of the time, and the lots that store it stand idle with it.[20] Parking is public infrastructure, built for one use and left empty for most of the day. The Powerhouse is the electric version of the parking lot. Its claim is that this ground can carry a civic or community amenity as well as cars.

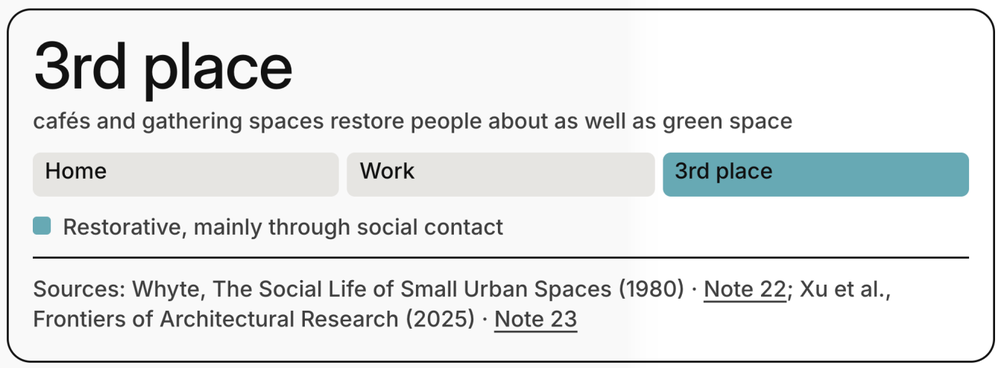
Space alone does not make a place. For more than sixty years New York has let developers build bigger in return for a plaza or an arcade at street level. When Jerold Kayden's team visited 503 of these spaces, they rated about two in every five as marginal.[21] William Whyte spent years filming people in plazas like these. He found that people stayed where there was something to do and somewhere to sit, and walked past everywhere else.[22] What holds people is the programme. A 2025 study monitored young adults' brain activity and stress. It found that third places, the cafés and gathering spaces between home and work, had restorative effects comparable to green space, working mainly through social contact.[23]

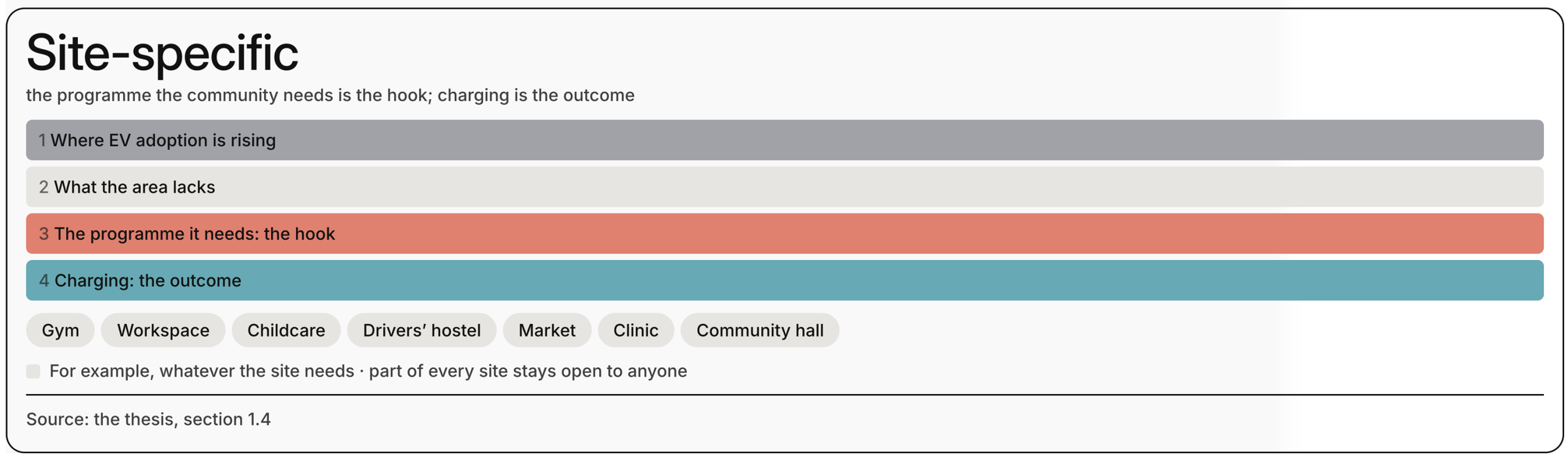
That is why the Powerhouse has no fixed brief. Each one is entirely site-specific: placed where electric vehicle adoption is rising, and shaped by what that area lacks. Its programme might be a gym, a clinic, a drivers' hostel or a market. Whatever it is, it is chosen for the people who live, work and travel nearby, and part of every site stays open to anyone.

For the same reason, the Powerhouse is not only for people who charge. The programme the community needs is the hook: it is what brings people to the site, on foot or by car. Charging is the outcome. It anchors the site, but it is not why people come. Seen this way, the Powerhouse is a first step towards a wider claim: that any parking lot or piece of dead space could be turned towards a programme that serves its area.









1.5 The Anchor

An anchor gives a place its reason to exist. The shopping mall made the idea explicit. When Victor Gruen designed Southdale outside Minneapolis in 1956, he hoped to give a new suburb the centre it lacked.[24] Two department stores at either end pulled shoppers through, and the rent from the smaller shops between them paid for the covered court where people met.[25] The anchor created the opportunity, and everything else grew around it.

At the Powerhouse the charger plays that part, though not by drawing the crowd. Charging needs a fixed stretch of time and a steady supply of energy, and it brings vehicles to the same place every day; that is what makes the site possible. The programme does the drawing. Paid programmes live off the people the site brings together, and their rent pays for energy generated on site, keeps charging and the free programmes free, and looks after the open ground. The mall's anchor was a department store and its public space was private property. The Powerhouse is anchored by a piece of public infrastructure, and the place it supports is built for the community.

1956

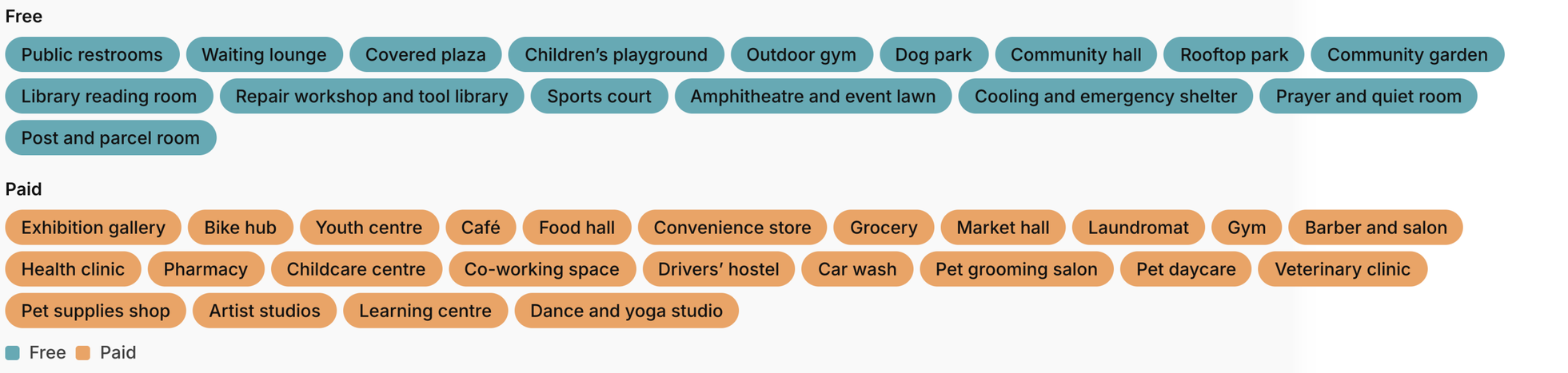
Southdale: two anchors pull people past the shops, whose rent pays for the court



Sources: Hardwick, Mall Maker (2004) · Note 24; Gruen and Smith, Shopping Towns USA (1960) · Note 25

40 programmes

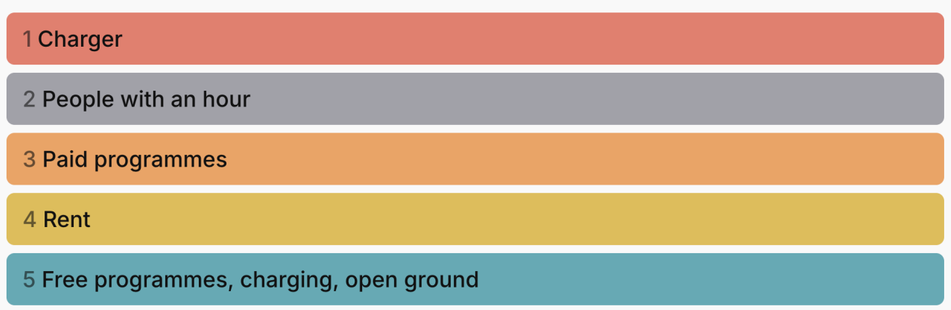
a stop can hold, free or paid; a site picks what its area lacks



Sorted with Jev (TypeSafe), judgements not survey data; unsure calls and four thesis choices set free. Recurring: FHWA NEVI guidance (restrooms, food service); Gilbarco Veeder-Root driver poll (2025); Deloitte driver study; Gridserve, Rove, Audi, IONNA, Love’s.

Free

charging, paid for by the rent of the programme it anchors

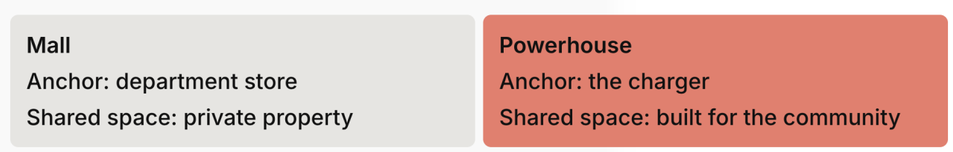


Each step pays for the next

Source: the thesis, section 1.5

Public

the Powerhouse is anchored by infrastructure, not a store



Source: the thesis, section 1.5

1.6 Argument, Objectives, and Scope

This thesis argues that an Electric Vehicle Powerhouse complex will turn charging activities into an opportunity for enhanced civic / community engagements. Placed where electric vehicle adoption is rising, each Powerhouse turns idle parking into a civic amenity with a programme the area lacks: the programme is the hook, and charging is the outcome. If the extra time leads only to more retail, the hypothesis fails.

Three questions test the argument, and each becomes a chapter. The first is about the stay: how long each kind of vehicle stops, from a quick charge to an overnight rest, and what that time allows. The second is about public space: how much of a roadside site is given to community use, measured as a share of site area and leaving out parking and charging bays. The third is about the programme: what each neighbourhood lacks, which programmes are free and which are paid, and whether together they give people a reason to stay.

The research starts in New York, where electric vehicle adoption is rising. It then widens to the rest of the United States, with precedents from the United Kingdom, Europe and Japan. Its ground is the parking lot and other dead space. It covers every road vehicle, and the people nearby who do not charge at all. The stay it plans for runs from a twenty-minute charge to a driver's overnight rest, by cable or wireless pad. Chargers at existing malls and supermarkets are set aside, because there shopping draws the charger rather than the other way round. Faster charging does not undo the argument. BYD's megawatt chargers promise hundreds of kilometres of range in five minutes,[26] but at the Powerhouse it is the programme, not the charge, that holds people. Few site plans are public and operators guard their finances, so the funding model is argued from municipal and community-energy precedents: the rent of paid programmes keeps charging and the free programmes free.

Argument

an Electric Vehicle Powerhouse turns charging into civic and community engagement

1 Parking: idle public infrastructure | 2 Its electric version, where EV adoption rises | 3 A programme the area lacks: the hook | 4 Charging: the outcome | 5 Civic and community life

Fails if the extra time leads only to more retail

Source: the thesis, sections 1.4–1.6

Objectives

three questions test the argument; each becomes a chapter

02 The Stop	Measure the stay: how long each vehicle stops, from a quick charge to an overnight rest
03 The Public Space	Measure the ground: the share of a roadside site given to community use, leaving out parking and charging bays
04 The Programme	Define the programme: what each neighbourhood lacks, free and paid, and whether it gives people a reason to stay

Source: the thesis, section 1.6

Scope

not a five-minute stop: any stay, any road vehicle, on ground parking already holds

Where	New York first, where EV adoption is rising and half of city car owners park on the street; then the US, with UK, European and Japanese precedents
Ground	Parking lots and other dead space
Who	E-bikes, motorcycles, cars, buses and trucks, and people who do not charge at all
Stay	From a 20-minute charge to an overnight rest, wired or wireless
Programme	Forty architectural programmes, free and paid, chosen for what the area lacks

Left out
Chargers at malls and supermarkets, where shopping draws the charger

Paid for by
The rent of paid programmes, which keeps charging and the free programmes free

Sources: the thesis, sections 1.1–1.5

1.7 Theoretical Framework

The thesis reads the Powerhouse through three lenses, each concerned with one part of its argument: sustainability with the Powerhouse and its charging, phenomenology with the stop, and social theory with the community and its public space.

The first lens is sustainability. Sustainable development is usually defined as meeting the needs of the present without compromising the ability of future generations to meet their own.[27] Here it has three parts. Environmentally, the Powerhouse builds on ground that parking already covers and makes and stores its own energy. Socially, it serves people who cannot charge at home and stays open to those who do not charge at all. Economically, its programme keeps the place running. Ian McHarg's practice of reading a site in layers, and Sim Van der Ryn and Stuart Cowan's principle that design should grow from its place and make natural systems visible, guide where a Powerhouse goes and how its energy is shown.[28]

The second lens is phenomenology: what it is like to spend the stop on site. Martin Heidegger argued that to build is to make dwelling possible, and that a built thing, like a bridge, gathers a place around it.[29] The charger can do the same. Christian Norberg-Schulz added that people need to know where they are and to feel that they belong,[30] which today's identical charging stops rarely allow. In these terms, the Powerhouse turns a pause in a journey into a place.

The third lens is social theory. Henri Lefebvre showed that space is produced: planned by those who design it and lived by those who use it.[31] Charging sites today are almost entirely planned, by standards and operators, for the vehicle. Lefebvre's right to the city,

the claim that everyone may use and shape urban space,[32] underlies the Powerhouse's promise of public space open to all. Michel Foucault's account of surveillance raises the opposite risk, that such places watch and move on the people who linger.[33] The Powerhouse answers with Jane Jacobs's alternative: safety that comes from people being present, not from being watched.[34]

3 lenses

three questions, answered in turn, lead to the Powerhouse

1 Sustainability	**Should it exist?**	Yes: it builds on land parking already covers, makes its own energy and serves people who cannot charge at home.
2 Phenomenology	**Can the stop become a place?**	Yes: the site gathers a pause in a journey into a place people know and belong to.
3 Social theory	**Who is it for?**	Everyone: lived by its users rather than planned for vehicles, and safe because people are there.
The Powerhouse	**A civic place on idle parking**	Open to all, sustained by its own energy and programme.

Source: the thesis, section 1.7

Each part

what the thinkers say, and what it means for the Powerhouse

1 Sustainability	Brundtland defines sustainable development as meeting today's needs without taking from future generations.[27] The Powerhouse meets it three ways: environmental (land parking already covers, its own energy), social (people who cannot charge at home) and economic (paid programmes keep it running). McHarg's layered reading of a site decides where a Powerhouse goes; Van der Ryn and Cowan ask that its energy be visible, not hidden.[28] *Backs 1.1 and 1.3.*
2 Phenomenology	Heidegger's bridge does not just cross a river; it gathers two banks into a place.[29] The site can do the same for a pause in a journey. Norberg-Schulz adds that people need to know where they are and feel they belong,[30] which today's identical charging stops, Augé's non-places, do not allow. *Backs 1.2 and 1.3.*
3 Social theory	Lefebvre separates space as planned from space as lived; charging sites today are planned by standards and operators for the vehicle.[31] His right to the city grounds the open ground and free programmes.[32] Foucault warns that such places can watch and move people on;[33] Jacobs answers that safety comes from people being present.[34] *Backs 1.4 and 1.5.*

Sources: notes 27–34

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The Stop

Parking, travel and the time to charge

Every way of travelling makes people wait, and every kind of wait has produced its own place. A rider who walks up to a bus stop waits about eight and a half minutes on average.[1] A driver at a gas station stays about five. A passenger at a large American airport spends a median of around eighty minutes in the terminal.[2] Each wait is sized to what the journey asks of it, and the building around it follows.

The private car is different. It is not waiting for its passengers; they are waiting to use it. In 2022 a typical American household vehicle stood parked for 95 per cent of the day and was driven for about sixty-five minutes.[3] That idle time is what makes the private electric car easy to charge. Most owners plug in at home and charge overnight while they sleep, and about four of every five charges happen this way.[4] For a driver with a driveway, the stop barely exists.

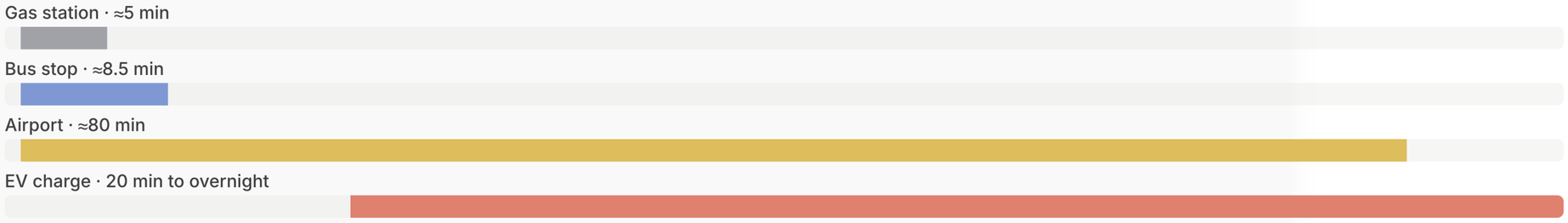
Not every vehicle can wait for night. Buses, taxis, ride-hail cars, delivery vans and trucks work all day. A full-time ride-hail car covers 180 to 190 miles a day,[5] against about fifty for a car in a one-car household,[6] and many ride-hail drivers live in apartments with no off-street parking. New York is electrifying exactly these fleets. The city's Taxi and Limousine Commission aims to make the vast majority of its licensed vehicles electric by 2030.[7] The MTA is committed to a zero-emission bus fleet by 2040, though its first electric buses could not get through a full day of service.[8] Trucks, as Chapter 1 showed, stop by law. These vehicles have to charge during the working day, away from home, and that is where the stop is made.

New York makes this case more sharply than anywhere else in the country. Across the United States, 69 per cent of workers drive to work alone; in New York State the figure is 49 per cent, the lowest of any state.[9] The rest ride buses, trains, taxis and ride-hail cars, or walk and cycle, and more of their journeys are made in vehicles that someone else owns and that never sleep in a driveway.

The public charging stop is therefore for four groups of people. The first is fleets charging mid-shift. The second is drivers without a driveway: half of New York City's car owners park on the street.[10] The third is travellers far from home. The fourth is the people nearby who come for the programme rather than the charge. This chapter is about the stop they share: what it is, how long it lasts, and what sets its length. It moves from the ground the stop occupies, to the network it belongs to, to the history of stops that became places, and then to the three things that set a stop's length today: the vehicle, the law and the speed of the charger. Of the three lenses set out in Chapter 1, it works mainly through two. Sustainability looks at the ground charging occupies and the energy it needs. Phenomenology looks at the stop as time that is lived, not just measured.[11]

5–80 min

every trip has a stop, and each way of travelling builds its own place to wait



Typical wait, on a 0–90 minute line · airport is the median dwell time at large US airports

Sources: Ohio State University, bus wait study (2020); Journal of Air Transport Management, passenger dwell time at US airports (2024); the thesis, Table 1.1

95%

of a typical day a household car stands parked; it is driven about 65 minutes



Parked · Driven · cars, unlike buses or trains, mostly wait

Source: US DOE, Fact of the Week #1356, from the 2022 National Household Travel Survey

4 in 5

charges happen at home, mostly overnight while the owner sleeps: the private car's daily routine

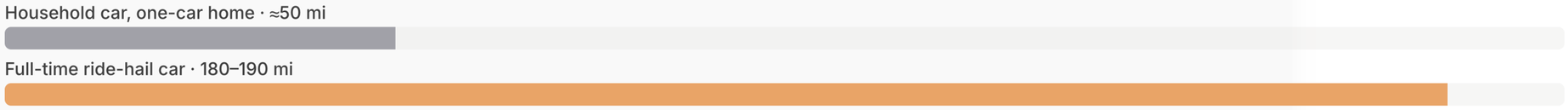


Home · Public and work · only for those with a driveway

Source: US DOE, "National EV Charging Network" (see 1.1)

185 mi

a full-time ride-hail car drives a day; fleets cannot wait for night and charge mid-shift



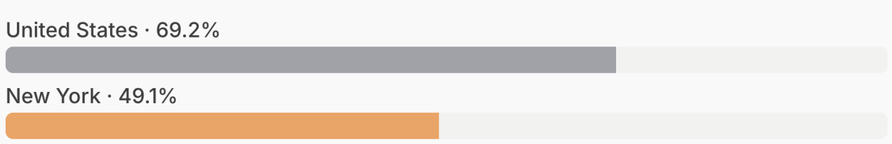
Buses	MTA: all 5,840 buses zero-emission by 2040; its first electric buses could not last a day of service
Taxis and ride-hail	TLC: the vast majority of licensed vehicles electric by 2030; 2 million EV rides by March 2024
Trucks	A 30-minute break after 8 hours at the wheel, 10 hours off at night

Miles driven a day · New York's public and hired fleets are going electric

Sources: US DOE, Fact of the Week #1332 (2024); ICCT, ride-hailing charging study (2020); Streetsblog NYC on the MTA audit (Aug 2026); NYC TLC, Charged Up! (2022) and Green Rides (2024); 49 C.F.R. § 395.3

49%

of New York workers drive to work alone, the lowest of any state (US: 69%)



The rest go by transit, taxi, ride-hail, shared car, bike or on foot: more shared vehicles to charge

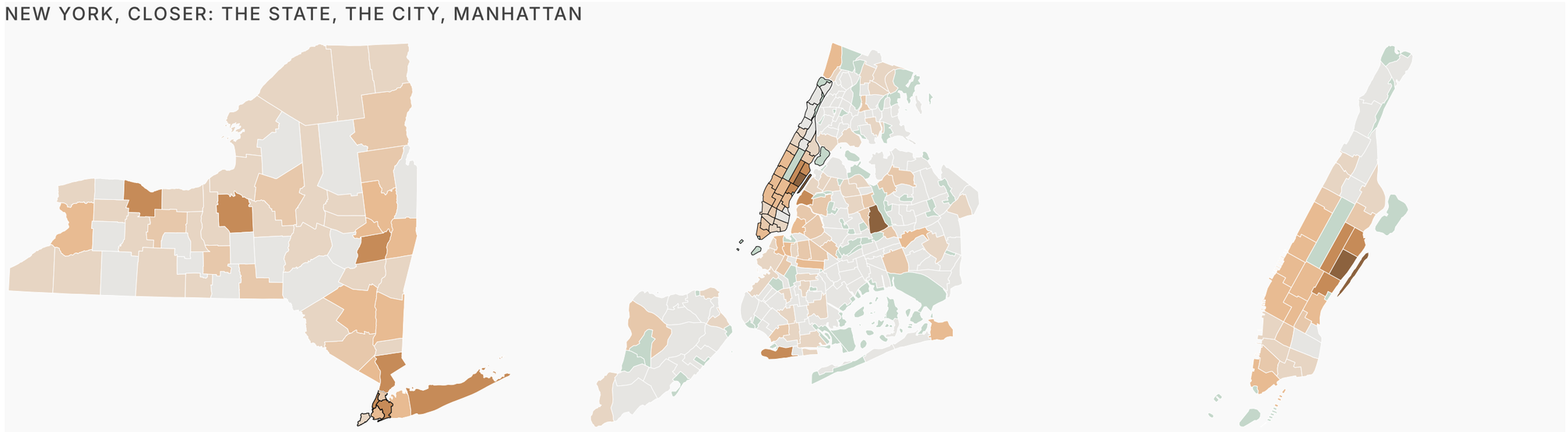
Source: US Census Bureau, American Community Survey 2023, via America's Health Rankings

4 groups

the people a public charging stop is for

Fleets	Buses, taxis, ride-hail, delivery and trucks, charging mid-shift
Drivers without a driveway	Half of New York City car owners park on the street
Travellers	Anyone a long way from home
Everyone else	People nearby who come for the programme, not the charge

Sources: NYC DOT; the cards above



2.1 A City Paved for Parking

American cities have covered a large share of their land in concrete, and most of it is for parking. Donald Shoup showed that free parking is paid for by everyone, in higher rents, higher prices and land that could have been used for something else.[12] The curb, the private lot, the gas station forecourt, the supermarket car park and the highway rest area are all versions of the same ground: paved space sized to the vehicle, not to the person in it.

New York shows what that ground does over a day. About 1.6 million people sleep in Manhattan; on a weekday about 3.1 million are there.[13] They come in by train and bus, and by road. Around 490,000 vehicles enter Manhattan below 60th Street each day,[14] carrying about 785,000 people in private cars,[15] while taxis and ride-hail cars make close to a million trips a day across the city.[16] The core's roughly 103,000 off-street parking spaces fill as they arrive.[17]

Each of those vehicles already stops. A commuter's car sits in a garage for eight or ten hours, long enough for a slow charge to fill it. A taxi or ride-hail car pauses between trips, long enough for a fast one. A bus waits at its layover, and a delivery van waits while it loads. As these vehicles go electric, the question is not where to build new stops but how to charge during the ones they already make, in the city, where they are. The Powerhouse does not replace the concrete. It is a layer added to it: paved ground that already exists, given a programme the area lacks, with charging as the outcome.

What matters, then, is the length of the stop. In this thesis it is defined as the time in minutes between a person arriving at a site and leaving it, on a visit whose main purpose is to charge a vehicle or to take a required rest. I call this the stop. Parking is the starting condition: the ground the stop already sits on, and the ground the Powerhouse proposes to give back. Reusing it, rather than paving new land, is the first sense in which the Powerhouse is sustainable: a solution that grows from the place it is in.[18]

2×

Manhattan's population on a weekday, as people come in by car, bus, train, taxi and ride-hail

At night, residents · ≈1.6 million

By day, with commuters · ≈3.1 million

Source: NYU Rudin Center, The Dynamic Population of Manhattan (Census daytime definition) · Note 13

490,000

vehicles a day enter Manhattan below 60th Street; about 785,000 people come in by private car

Sources: MTA congestion relief zone counts, via ABC7 (Jan 2025) · Note 14; NYMTC, Hub Bound Travel Data 2024 · Note 15

≈1 million

taxi and ride-hail trips a day across the city, by vehicles that work all day

Source: NYC TLC (243 million high-volume ride-hail trips in 2025, plus yellow taxis) · Note 16

103,000

off-street parking spaces in the Manhattan core, filled by the cars that come in each weekday

Sources: NYC Planning, Manhattan Core Public Parking Study (2011) · Note 17; DCWP licenses 1,600+ garages and lots citywide

5 grounds

one kind of space: paved, sized to the vehicle, used minutes at a time

Curb space · Private lot · Gas station forecourt · Supermarket car park · Highway rest area

Sources: Shoup, The High Cost of Free Parking (2005) · Note 12

Charge where it parks

the vehicles that fill the city by day could charge in it, during the stops they already make

Commuter car	Parked 8–10 hours in a garage: a slow charge fills it by evening
Taxi and ride-hail	Charges between trips, 20–60 minutes on a fast charger
Bus	Tops up at a layover or depot inside the city
Delivery van	Charges while it loads, or overnight at the curb

Sources: the cards above; US DOT, "Charger Types and Speeds"

2.2 The Infrastructure of Vehicle Travel

Charging is not one place but a network, and each part of it produces a different kind of stop. It helps to read it as an anatomy, from the slowest charger to the fastest.

Home and workplace. Most charging happens where the vehicle is already parked for hours. A car plugged into a wall socket or a home charger fills slowly overnight, and an office car park does the same during the working day. By the Department of Energy's estimate, about four in five charges happen at home.[19] These charges create no stop at all, because the vehicle was going to be standing there anyway.

Destination charging. Slower public chargers at supermarkets, hotels, car parks and on the street let people top up while they do something else. The stop is set by the errand, not by the charge.

Fast charging. Along highways and in city hubs, fast chargers fill a car in well under an hour. This is where the stop is created by the charge itself. The driver is there because the car needs energy, and has to wait for it. It is the closest thing to the gas station, and the clearest case for the Powerhouse.

Freight and fleet charging. Trucks, buses and delivery vans charge at depots overnight or at high-power chargers along freight routes. Freight already has a stop it cannot

avoid: drivers are required by law to rest, and the places to do so are short. The lack of truck parking is one of the most pressing problems in American trucking, and drivers near the end of their legal hours keep driving to look for a space.[20]

Battery swapping. For scooters, and for some cars and trucks, the empty battery can be taken out and replaced with a full one. The vehicle stops for minutes, but the battery it leaves behind still has to be charged somewhere.

This anatomy creates the first problem for measuring the stop. Because most charging happens at home, the people at public chargers are a particular group: travellers on long trips, people who live in apartments or park on the street, ride-hail drivers and fleets. They are not a cross-section of all drivers, and a study of them alone could give a false picture. This is a problem of selection bias. The study responds by choosing sites that serve different users, such as a highway site used by travellers and trucks, a city hub used by residents and ride-hail drivers, and a small-town site used by local people, and by asking each visitor why they are there. It also counts people who arrive without a vehicle, since the Powerhouse is meant to be open to everyone.

2.3 Stations, Airports and Service Areas

The link between the stop and the making of a place is old, and Chapter 1 traced it from Charles Horton Cooley's observation of 1894 that "population and wealth tend to collect wherever there is a break in transportation."[21] What matters here is how differently each type treated the person. The railway station and the airport were built around the wait, filling it with halls, shops and restaurants.[22] The gas station went the other way, and it barely changed for a century.[23] It was built for the vehicle, and the person was expected to leave as soon as possible.

The Powerhouse proposes a new type in this line: a station for electric road transport. Like the railway station, it is organised around a break in the journey. Unlike the gas station, it is designed for the person as well as the vehicle. It also has a second purpose. By making charging somewhere people want to be, it makes the switch to electric vehicles easier to choose. In phenomenological terms, it turns a pause in a journey into a place: somewhere to dwell for the length of the stop, rather than a queue to get through.[24]

This history suggests the method. The study does not control the length of the stop; nobody can make a car charge more slowly to see what happens. It is comparative instead. It selects existing sites where the stop is short, medium, long and overnight, and compares how much of each site is given to people. This follows the building-type studies of Pevsner and Geist introduced in Chapter 1: follow one type through many examples and look for what changes with the conditions.[25] The stop is therefore measured rather than manipulated, and the sites are chosen to cover its full range.

The history also warns of the most serious problem. The hypothesis says a longer stop leads to more public space, but the reverse can also be true. People stay longer at stations and service areas that give them somewhere to sit and something to eat, and truck drivers choose rest areas with better amenities and safety features.[26] If the study only observed that sites with more public space have longer stops, it could not tell which caused which. This is a threat to internal validity. The response is to take the length of the stop from the charger, the vehicle and the law, which are fixed before anyone decides what to build around them, and to use the time people actually stay as a check.

2.4 Vehicle Size and Charging Time

The length of the stop depends first on what arrives. Charging time is set by two numbers: how much energy the battery holds, measured in kilowatt-hours (kWh), and how fast the charger can deliver it, measured in kilowatts (kW). Dividing the first by the second gives a rough time. A 75 kWh car battery charged from 10 to 80 percent needs about 52 kWh. At 100 kW that takes about half an hour; at 50 kW, about an hour; on a 7 kW home charger, most of a working day.[27] Small vehicles need little energy and can charge from a wall socket. Trucks sit at the far end. Anderson and Kish note that charging an electric truck can take up to eight hours, while refuelling a diesel truck takes "a mere five to ten minutes."[28] Table 2.1 sets out the range.

Table 2.1. Vehicle size and charging time.

Vehicle	Typical battery	Typical charger	Time, 10 to 80 percent	Stop
Electric bicycle	0.4–0.7 kWh	Wall socket, under 1 kW	2–4 hours; or swap in minutes	Short (swap) or none (home)
Scooter, motorcycle	2–15 kWh	1–3 kW socket; some fast charging	1–5 hours; or swap in minutes	Short (swap) or long
Car	40–100 kWh	7–11 kW (slow); 50–350 kW (fast)	15–60 minutes fast; 4–10 hours slow	Medium
Pickup, large SUV, van	100–200 kWh	50–350 kW fast	30–90 minutes fast	Medium to long
Bus, coach	300–600 kWh	150–600 kW depot or route	1–3 hours	Long
Heavy truck	400–1,000 kWh	350 kW–1 MW and above	30 minutes–2 hours fast; up to 8 hours slow	Long or overnight

Battery and charger figures are approximate ranges from manufacturers' published specifications for current models. Times are the author's calculation (energy needed divided by average charging power) and are lower limits, because charging slows as the battery fills.

The table groups the stop into four lengths. A short stop is under fifteen minutes. A medium stop is fifteen minutes to an hour, which is a car at a fast charger. A long stop is one to three hours, typical of buses, large vehicles and trucks at fast chargers. An overnight stop is eight hours or more. These lengths match the lengths of stay used in Chapter 4 to choose a programme for the site.

Charge time, however, is not the same as waiting time. A car may need forty minutes of charge, but the driver may leave after twenty because they have enough to get home, or stay long after it finishes because they are eating. Some drivers never leave the car and spend the stop on their phones. If the study measured only the charge, it would describe the machine, not the person. This is a problem of construct validity.

The response is simple. The study records both: the time the vehicle needs to charge, worked out from Table 2.1, and the time people actually stay, observed on site using the timed counts developed by William Whyte and by Jan Gehl and Birgitte Svarre.[29] The difference between the two is the point. A charging site that only holds people for as long as the charge takes is still a forecourt. The Powerhouse aims for people to stay longer than they have to, because the place is worth staying in. Observation is kept consistent by visiting each site at the same times of day, over several days, and nobody is identified.

2.5 Regulation: Driving Breaks, Charger Standards and Zoning

For professional drivers, the length of the stop is set partly by law rather than technology. In the United States a truck driver may drive no more than eleven hours after ten consecutive hours off duty, and must take a thirty-minute break after eight hours of driving.[30] In the European Union a driver of a truck or coach must take a break of at least forty-five minutes after four and a half hours at the wheel, and a daily rest of at least eleven hours, which may be shortened to nine.[31] Table 2.2 places these rules in the same four lengths as the vehicles.

Table 2.2. Required breaks for drivers.

Rule	Who	Required break	Stop
No legal rule	Private car drivers	None; set by the charge and the driver	Short to medium
US driving break	Truck drivers	30 minutes after 8 hours of driving	Medium
EU driving break	Truck and coach drivers	45 minutes after 4.5 hours of driving	Medium
US daily rest	Truck drivers	10 consecutive hours off duty	Overnight
EU daily rest	Truck and coach drivers	11 hours, or 9 hours when reduced	Overnight

These rules apply however fast the truck charges. A required rest that coincides with a required charge is a long stop that is very likely to happen, which makes freight the most stable case for this study and the first one it tests.

Charger standards set the stop for cars. The federal National Electric Vehicle Infrastructure standards require funded highway stations to offer at least four fast-charging ports, each able to deliver 150 kilowatts at the same time.[32] That figure is a practical baseline for highway stops over the next decade, because it is what public money is building, and it is the figure used to calculate the charge time at highway sites.

Zoning decides what may be built around the charger. Local codes were largely written for gas stations and parking lots, and they shape whether a charging site can include food, seating, workspace or rooms to stay. The Powerhouse does not fit those categories. It is not a plaza added to a car park; it works like a third place with charging at its centre, which is a different way of using the ground altogether. Its programme comes from the needs of the area around it. On a highway or freight corridor it might be somewhere to stay, with rooms and showers for drivers on their required rest. In a city it might be a gym or a place to work. In a town without one, it might be a medical centre.

This affects how much of the site is given to people more than the length of the stop itself, and it is taken up in Chapters 3 and 4. For this chapter, zoning matters because two sites with the same stop may give very different amounts of ground to people simply because one is allowed to and the other is not. The study records the zoning of each site so that its effect is not mistaken for the effect of the stop.

2.6 Faster Charging and the Future of the Stop

The stop is a moving target. Charging is getting faster. In 2025 BYD announced a one-megawatt "flash charging" system that it claims adds around 400 kilometres of range in five minutes.[33] Battery swapping points the same way, replacing a whole battery in minutes. If these spread, the car stop could fall back towards the length of a fuel stop within the life of the buildings this thesis proposes. Cooley saw something similar with the railways: as transfers improved, small breaks disappeared. In his words, "the number of breaks represents the expense of transportation."[34]

That does not undo the thesis. A faster stop for the driver does not mean less charging overall. The energy still has to come from somewhere, and a battery swapped out of a vehicle in five minutes still needs hours to recharge. Flash charging at a megawatt draws on the grid far harder than a slow charger, and needs storage and supply close by. The Powerhouse can be that place: the swap centre where batteries are held and recharged, and the local source of power, generating and storing its own electricity for the vehicles around it. It is meant to be a powerhouse in the literal sense, solving transport's need for electricity as well as the person's need for somewhere to be. This is the sustainability lens at its most literal: the site produces energy as well as using it.[35]

Stan Allen's account of "infrastructural urbanism" explains how a building prepares for that shift: infrastructure works as a field, a strong structural frame that allows uses no one has planned yet, so the structure stays while what happens on its floors changes over time.[36] A multi-storey garage that holds five levels of parked cars today can open with a public ground floor and charging decks above, and later—if charging speeds up or autonomous fleets need fewer stalls—shift its upper floors from vehicles to energy storage, community programme and a solar public roof (Table 2.3).

Table 2.3. Infrastructure as a field for the future: build the frame, let the programme change (after Stan Allen, 1999).

Floor level	Today (five floors of parked cars)	Opening (charging & civic base)	Later (fewer cars, more programme)
Roof / Level 5	Parked cars	Parked cars / solar canopy	Public roof & solar generation
Level 4	Parked cars	Parked cars	Community programme
Level 3	Parked cars	EV charging	Community programme
Level 2	Parked cars	EV charging	Fast charging & battery storage
Ground floor	Parked cars	Public ground floor	Civic station hall & public room

This also changes what the stop depends on. If the charge no longer forces people to stay, they will stay only where the site gives them a reason, which is the role of the programme, and why the hypothesis does not rest on charging time alone. The study allows for this in three ways. It puts most weight on the stops that technology does not control, such as the required rest for drivers, which will still apply when a truck can charge in minutes. It uses the four lengths of stop rather than exact minutes, so a site only has to stay within the same range. And it asks which parts of the programme would still work if the stop were halved. Because the Powerhouse is a charging hub, energy store and third place at once, it can adapt as vehicles and charging change, rather than becoming obsolete with them.

Summary

The stop is where this thesis begins: the time a person spends at a site whose main purpose is to charge a vehicle or take a required rest. It sits on ground that parking already occupies, within a network where most charging happens at home, and it continues a long history in which the break in a journey became a place. Its length is set by the vehicle, the law and the speed of the charger, and it ranges from a few minutes to most of a day. The study compares existing sites across four lengths of stop, recording both the time the charge needs and the time people actually stay.

The main challenges are that public-charger users are not typical drivers, that a good site can lengthen the stop as well as respond to it, that charge time is not the same as time on site, that zoning can shape a site as much as the stop does, and that faster charging may shorten the stop. The responses are to study different kinds of sites and users, to take the stop from the charger, vehicle and law and use observation as a check, to record the person as well as the plug, to note each site's zoning, and to rest the argument on required breaks and ranges of stop rather than on charger speed. Even as charging gets faster, the Powerhouse remains a source of power for transport and a place worth stopping in. Read through sustainability, the stop sits on reused ground and draws on energy made on site; read through phenomenology, it is time that can be lived rather than waited out. Chapter 3 turns to that place, through phenomenology and social theory.

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